

Word Problems & Reasoning



Number and digit reasoning

- 1 A number decreased by 9 equals 34. Find the number.
 - 2 Three times a number, minus 7, equals 32. Find the number.
 - 3 The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.
 - 4 A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.
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Age problems

- 5 A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.
 - 6 Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.
 - 7 Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.
 - 8 The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.
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Distance, rate and time

- 9 A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?
 - 10 Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?
 - 11 Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?
 - 12 A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?
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Work-rate problems

- 13 Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?
- 14 A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?
- 15 6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?
- 16 A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?
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Mixture and money problems

- 17 A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?
- 18 A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?
- 19 A solution is 40% acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to 25% acid?
- 20 Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?
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Logical deduction

- 21 Ali, Basim and Khalid have 36 books together. Ali has twice as many as Basim, and Khalid has 6 more than Ali. Find how many books each has.
- 22 A number between 50 and 70 is divisible by both 3 and 4, but not by 8. Find the number.
- 23 In a class, every student plays football or basketball (or both). 25 play football, 18 play basketball, and 10 play both. How many students are in the class?
- 24 Four consecutive even integers sum to 92. Find the largest one.
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Working backward

- 25 After spending half his money on a book, then 20 riyals on lunch, Yousef has 30 riyals left. How much did he start with?

- 26 A number is tripled, then 8 is subtracted, giving 25. Find the original number.
- 27 After a 15% pay cut, an employee's salary is 3,400 riyals. Find the salary before the cut.
- 28 A water tank is $\frac{1}{4}$ full. After adding 60 liters, it becomes $\frac{5}{8}$ full. Find the tank's total capacity.
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Mixed word problems

- 29 A rectangular field is twice as long as it is wide, and its perimeter is 90 m. Find its length and width.
- 30 Layla buys 3 notebooks and 2 pens for 29 riyals. Fahad buys 1 notebook and 4 pens for 23 riyals, at the same prices. Find the price of one notebook and one pen.
- 31 A car rental costs 80 riyals per day plus 0.50 riyals per km driven. A 3-day rental cost 435 riyals in total. How many km were driven?
- 32 The sum of three consecutive odd integers is 63. Find the integers.
- 33 A shop sells notebooks for 6 riyals each, with a "buy 3, get 1 free" offer. Layla needs 20 notebooks. How much will she pay?
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More age and number problems

- 34 The sum of a number and its square is 42. Find the positive value of the number.
- 35 A number's triple exceeds its double by 15. Find the number.
- 36 In 10 years, Noura will be 3 times as old as she was 2 years ago. Find her current age.
- 37 The sum of two consecutive multiples of 5 is 75. Find the larger one.
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More rate and proportion problems

- 38 A photocopier prints 45 pages per minute. How many minutes does it take to print 810 pages?
- 39 A recipe for 6 people uses 450 g of rice. How much rice is needed for 15 people, at the same rate per person?

- 40 A factory produces 1,200 items in 8 hours. At the same rate, how many items would it produce in a 10-hour shift?
- 41 Three taps together fill a pool in 4 hours. Two of the taps alone would take 6 hours. How long would the third tap alone take to fill the pool?
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MCQ — Number and digit reasoning

- 42 A number decreased by 12 equals 47. Find the number.
- a. 59
 - b. 65
 - c. 35
 - d. 47
- 43 Four times a number, minus 9, equals 47. Find the number.
- a. 56
 - b. 14
 - c. 11
 - d. 9.5
- 44 The sum of two numbers is 74. One number is 18 more than the other. Find the larger number.
- a. 46
 - b. 37
 - c. 28
 - d. 56

- 45 A two-digit number has digit sum 9. Reversing its digits gives a number 27 more than the original. Find the original number.
- a. 18
 - b. 36
 - c. 63
 - d. 45
- 46 A number is increased by 25% to give 60. Find the original number.
- a. 75
 - b. 40
 - c. 45
 - d. 48
- 47 A number plus twice itself equals 27. Find the number.
- a. 13.5
 - b. 27
 - c. 9
 - d. 18

MCQ — Age problems

- 48 A father is 5 times as old as his son. In 4 years, he will be 3 times as old as his son will be. Find the father's current age.
- a. 20
 - b. 16
 - c. 24
 - d. 4

- 49** Huda is twice as old as her sister. In 8 years, the sum of their ages will be 46. Find Huda's current age.
- a. 10
 - b. 18
 - c. 14
 - d. 20
- 50** Khalid is 6 years older than Faisal. Five years ago, Khalid was twice as old as Faisal. Find their current ages.
- a. Faisal = 17, Khalid = 11
 - b. Faisal = 11, Khalid = 6
 - c. Faisal = 5, Khalid = 11
 - d. Faisal = 11, Khalid = 17
- 51** The sum of a mother's and son's ages is 54. The mother is 24 years older than the son. Find the son's age.
- a. 30
 - b. 15
 - c. 27
 - d. 39
- 52** In 8 years, Mona will be twice as old as she was 4 years ago. Find her current age.
- a. 8
 - b. 20
 - c. 16
 - d. 12

53 Tariq is currently twice as old as his son. If Tariq is 40, find his son's age.

- a. 18
- b. 20
- c. 22
- d. 80

MCQ — Distance, rate and time

54 A train travels 210 km in 3 hours. At the same speed, how far does it travel in 7 hours?

- a. 490
- b. 420
- c. 70
- d. 630

55 Two cities are 420 km apart. A car leaves one city for the other at 60 km/h. How long does the trip take?

- a. 6 hours
- b. 25,200 hours
- c. 7 hours
- d. 8 hours

56 Two cyclists start at the same point and ride in opposite directions, at 12 km/h and 18 km/h. After 3 hours, how far apart are they?

- a. 54
- b. 90
- c. 30
- d. 36

- 57 A plane flies against a headwind at 400 km/h and covers 1,600 km. How many hours does the flight take?
- a. 4,000
 - b. 4
 - c. 5
 - d. 400
- 58 A car travels at 90 km/h for 2.5 hours. Find the distance covered.
- a. 225
 - b. 36
 - c. 92.5
 - d. 180
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MCQ — Work-rate problems

- 59 Worker A can paint a room in 8 hours; Worker B can paint the same room in 4 hours. Working together, how long will it take?
- a. 4 hours
 - b. 12 hours
 - c. 6 hours
 - d. $\frac{8}{3}$ hours
- 60 A pipe fills a tank in 12 hours; a second pipe fills it in 6 hours. Working together, how long to fill the tank?
- a. 3 hours
 - b. 18 hours
 - c. 4 hours
 - d. 9 hours

- 61 8 workers can complete a job in 6 days. How many days would 12 workers take, at the same rate?
- a. 4
 - b. 9
 - c. 3
 - d. 6
- 62 A pipe fills a tank in 10 hours, but a leak can empty a full tank in 20 hours. If both act at once, how long to fill the tank?
- a. 20 hours
 - b. 30 hours
 - c. 10 hours
 - d. 15 hours
- 63 Worker C can complete a task in 9 hours. After working alone for 3 hours, what fraction of the task remains?
- a. $\frac{2}{3}$
 - b. $\frac{1}{3}$
 - c. $\frac{1}{6}$
 - d. $\frac{1}{9}$

MCQ — Mixture and money problems

- 64 A jeweler mixes gold worth 250 riyals/gram with silver worth 40 riyals/gram to make 12 grams of a mixture worth 145 riyals/gram. How many grams of gold are used?
- a. 6
 - b. 4
 - c. 8
 - d. 9
- 65 A shopkeeper has 21 riyals in 1-riyal and 2-riyal coins, 15 coins in total. How many 2-riyal coins does he have?
- a. 9
 - b. 15
 - c. 6
 - d. 3
- 66 A solution is 30% acid. How many liters of pure water must be added to 15 liters of this solution to make it 20% acid?
- a. 4.5
 - b. 7.5
 - c. 5
 - d. 10
- 67 Event tickets cost 25 riyals for adults and 15 riyals for children. 200 tickets were sold for 4,300 riyals. How many adult tickets were sold?
- a. 150
 - b. 130
 - c. 100
 - d. 70

- 68 A syrup is 25% sugar. How many kilograms of pure sugar must be added to 8 kg of this syrup to make it 40% sugar?
- a. 3.2
 - b. 4
 - c. 1
 - d. 2
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MCQ — Logical deduction

- 69 Omar, Bandar and Nasser have 44 books together. Omar has twice as many as Bandar, and Nasser has 8 more than Bandar. Find how many books Bandar has.
- a. 22
 - b. 17
 - c. 9
 - d. 18
- 70 A number between 80 and 100 is divisible by both 4 and 6, but not by 8. Find the number.
- a. 84
 - b. 96
 - c. 88
 - d. 90

- 71 In a class, every student plays chess or checkers (or both). 20 play chess, 15 play checkers, and 8 play both. How many students are in the class?
- a. 27
 - b. 35
 - c. 12
 - d. 43
- 72 Four consecutive odd integers sum to 96. Find the largest one.
- a. 30
 - b. 27
 - c. 24
 - d. 21
- 73 Three consecutive integers sum to 72. Find the smallest one.
- a. 25
 - b. 24
 - c. 23
 - d. 22

MCQ — Working backward

- 74 After spending half his money on a book, then 15 riyals on lunch, Faisal has 25 riyals left. How much did he start with?
- a. 100
 - b. 65
 - c. 80
 - d. 40

- 75** A number is doubled, then 6 is subtracted, giving 20. Find the original number.
- a. 16
 - b. 7
 - c. 26
 - d. 13
- 76** After a 20% pay cut, an employee's salary is 3,200 riyals. Find the salary before the cut.
- a. 4,200
 - b. 4,000
 - c. 3,600
 - d. 3,840
- 77** A water tank is $\frac{1}{3}$ full. After adding 80 liters, it becomes $\frac{3}{4}$ full. Find the tank's total capacity.
- a. 240
 - b. 192
 - c. 120
 - d. 106.7
- 78** A number is halved, then 10 is added, giving 34. Find the original number.
- a. 48
 - b. 44
 - c. 88
 - d. 24

MCQ — Mixed word problems

- 79 A rectangular field is twice as long as it is wide, and its perimeter is 108 m. Find its length and width.
- a. Length = 36, width = 36
 - b. Length = 18, width = 36
 - c. Length = 54, width = 27
 - d. Length = 36, width = 18
- 80 Sarah buys 4 notebooks and 3 pens for 38 riyals. Khalid buys 2 notebooks and 5 pens for 40 riyals, at the same prices. Find the price of one notebook.
- a. 6
 - b. 8
 - c. 5
 - d. 4
- 81 A car rental costs 100 riyals per day plus 0.50 riyals per km driven. A 4-day rental cost 500 riyals. How many km were driven?
- a. 250
 - b. 1,000
 - c. 200
 - d. 100
- 82 The sum of three consecutive even integers is 78. Find the largest one.
- a. 28
 - b. 26
 - c. 30
 - d. 24

- 83 A shop sells pens for 4 riyals each, with a "buy 4, get 1 free" offer. Karim needs 25 pens. How much will he pay?
- a. 100
 - b. 64
 - c. 80
 - d. 96
- 84 A cyclist travels 60 km at a certain speed, then returns the same 60 km at half that speed, taking 9 hours total. Find the original speed.
- a. 15
 - b. 10
 - c. 40
 - d. 20
- 85 Two numbers have a ratio of 3 : 5 and a sum of 96. Find the larger number.
- a. 60
 - b. 48
 - c. 36
 - d. 72

MCQ — More rate and proportion problems

- 86 A photocopier prints 60 pages per minute. How many minutes does it take to print 900 pages?
- a. 18
 - b. 12
 - c. 54,000
 - d. 15

- 87** A recipe for 8 people uses 600 g of rice. How much rice is needed for 20 people, at the same rate per person?
- a. 750
 - b. 1,500
 - c. 1,800
 - d. 1,200
- 88** A factory produces 1,500 items in 6 hours. At the same rate, how many items would it produce in a 9-hour shift?
- a. 1,875
 - b. 2,700
 - c. 2,250
 - d. 2,000
- 89** Three taps together fill a pool in 5 hours. Two of the taps alone would take 8 hours. How long would the third tap alone take?
- a. 3 hours
 - b. 13 hours
 - c. $\frac{40}{3}$ hours
 - d. 40 hours
- 90** A car uses 8 liters of fuel to travel 96 km. At the same rate, how many liters are needed to travel 156 km?
- a. 13
 - b. 14.6
 - c. 19.5
 - d. 12

91 A map has a scale of 1 : 50,000. If two towns are 6.4 cm apart on the map, find the actual distance in km.

a. 3.2

b. 0.32

c. 320

d. 32